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Advertising Expensive Mortgages

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1 Big picture

- **Question.** Does mortgage advertising help borrowers find cheaper mortgages, or does it persuade them into more expensive ones?
- **Setting.** The paper studies the U.S. subprime mortgage market using matched data on local advertising by mortgage lenders and loan-level mortgage contracts, with a special focus on adjustable-rate mortgages (ARMs).
- **Core challenge.** A positive correlation between advertising and mortgage prices could reflect persuasion, but it could also come from local housing booms, lender characteristics, borrower risk, or higher costs of serving certain borrowers.
- **Main conceptual contribution.** The paper does not look only at whether advertising increases mortgage demand. It asks whether advertising changes the *quality of the mortgage choice* by steering consumers toward relatively worse mortgages.
- **Why ARMs matter.** The main object is the ARM reset rate. Introductory teaser rates are salient and often advertised, while reset rates are the hidden or less salient dimension that determines how expensive the mortgage later becomes.
- **Main empirical design.** The authors first construct a loan-level measure of mortgage expensiveness as the residual reset rate after controlling for borrower, contract, location, and time characteristics, including the initial rate. They then aggregate this to the lender-DMA-quarter level and regress lender expensiveness on local advertising intensity with lender, region, and quarter fixed effects.
- **Main baseline result.** Lenders that advertise more within a region charge more expensive mortgages in that region. In the baseline ARM specification, the coefficient on advertising is positive and significant, and the paper interprets a one-standard-deviation increase in advertising as raising reset rates by roughly 80 basis points.
- **Main heterogeneity result.** The positive advertising-expensiveness relation is concentrated in areas with more minority, less educated, and poorer borrowers, consistent with greater susceptibility among less sophisticated consumers.
- **Main credibility result.** The results survive multiple tests against alternative explanations. Advertisers' borrowers are not more delinquent; direct evidence from a large subprime lender suggests catering costs are too small to explain the magnitudes; and a Craigslist-based instrument that shifts local advertising also produces a positive effect of advertising on mortgage expensiveness.
- **Main content result.** In 37,432 print and direct-mail campaigns, only seven explicitly mention a reset rate. Initial or teaser rates are salient and frequent; reset rates are almost never disclosed. Advertised APRs do not reliably lead borrowers to cheaper lenders.
- **Economic takeaway.** The evidence strongly supports the persuasion view of mortgage advertising: advertising did not mainly help consumers search better; it helped steer some borrowers into worse mortgage contracts.

2 Data and measurement (key objects)

2.1 Institutional setting and why the paper studies subprime ARMs

- The paper is motivated by the post-crisis debate over whether naive borrowers were misled by mortgage advertising into costly products.
- It focuses on the subprime market, where teaser-rate ARMs and later reset rates were at the center of lawsuits, policy scrutiny, and consumer-protection reforms.
- The key institutional idea is simple: if the initial rate is salient but the reset rate is shrouded, advertising may attract attention precisely to the wrong margin.

Interpretation. The paper is not studying advertising in a generic product market. It is studying a setting where the advertised feature and the economically important feature can differ sharply.

2.2 Advertising data: TNS Media Intelligence

- Advertising data come from TNS Media Intelligence (TNSMI).
- A record in the data is the amount a lender spends in a given month, media channel, and designated market area (DMA).
- The main channels are local newspapers, spot TV, outdoor advertising, radio, magazines, and other local media.
- The empirical analysis aggregates these data to the lender-DMA-quarter level to match the mortgage data.

The matched advertising sample covers 2002 to 2006.

Interpretation. DMAs are the natural advertising markets. Using lender-DMA-quarter variation is central because it lets the paper compare the same lender across different local advertising environments.

2.3 Mortgage data: LoanPerformance

- Mortgage data come from LoanPerformance, a large loan-level database covering the nonagency securities market.
- The paper keeps purchase mortgages only, for owner-occupied single-family homes, townhouses, and condominiums.
- It excludes FHA/VA-insured and other nonconventional loans, buy-down mortgages, and loans without valid FICO scores.

The final matched sample links 571 lender names from TNSMI to LoanPerformance lenders. It covers 105 of the 206 DMAs, corresponding to about 92% of the U.S. population.

Interpretation. The identifying data structure is unusually rich: local advertising by lender, loan-level contract terms, borrower characteristics, and later loan performance.

2.4 Key loan samples and descriptive statistics

Table I reports the main summary statistics.

For ARM loans:

- mean reset rate: 8.59%,
- mean initial rate: 7.8%,
- mean reset time: 28.24 months,
- mean LTV: 84.68,
- mean loan amount: about \$168,601,
- mean FICO: 626,
- prepayment-penalty share: 0.77,
- low-documentation share: 0.40,
- observations: 1,182,080.

For FRM loans:

- mean interest rate: 9.33%,
- mean LTV: 91.55,
- mean loan amount: about \$104,228,
- mean FICO: 640,
- observations: 464,530.

Interpretation. The subprime sample is economically meaningful and large, with substantial variation in pricing, borrower quality, and contract terms.

2.5 Local advertising intensity

The paper emphasizes that more than 96% of DMA-level mortgage advertising expenditures flow through three outlets: local newspapers, spot TV, and outdoor advertising.

Newspapers dominate local mortgage advertising, accounting for about 70% of total expenditures, versus roughly 14% each for spot TV and outdoor advertising.

Interpretation. This matters because if one particular channel is driving the pricing effect, that channel is most likely to be newspapers, where mortgage rate language can be displayed most directly and where Craigslist later becomes a close substitute.

2.6 Mortgage expensiveness: loan level

For ARM loans, the paper defines mortgage expensiveness from the residual of

$$y_{ijlt} = \beta i_{ijlt} + \alpha_t + \alpha_l + \Gamma X_{ilt} + \varepsilon_{ijlt}, \quad (1)$$

where:

- y_{ijlt} is the ARM reset rate for loan i issued by lender j in market l and quarter t ,
- i_{ijlt} is the initial ARM rate,
- X_{ilt} includes borrower and contract characteristics such as LTV, FICO, prepayment-penalty status, documentation status, and local demographic controls,
- α_t are quarter fixed effects,
- α_l are market fixed effects.

The paper then defines loan-level expensiveness as

$$\hat{\varepsilon}_{ijlt} = y_{ijlt} - \left(\hat{\beta} i_{ijlt} + \hat{\alpha}_t + \hat{\alpha}_l + \hat{\Gamma} X_{ilt} \right). \quad (2)$$

Interpretation. If two observably identical borrowers in the same market and quarter receive the same type of mortgage and the same initial rate, but one gets a higher reset rate, that borrower made the worse mortgage choice. The residual is the paper's measure of that relative expensiveness.

2.7 Mortgage expensiveness: lender level

Because advertising is observed at the lender-DMA-quarter level, the paper aggregates loan-level residuals to form lender expensiveness:

$$p_{jlt} = \frac{1}{n_{jlt}} \sum_i \hat{\varepsilon}_{ijlt}, \quad (3)$$

where n_{jlt} is the number of loans by lender j in DMA l and quarter t .

The mean gap between the 95th and 5th percentile lenders in the same DMA-quarter is about 2.8 percentage points.

Interpretation. This aggregation removes idiosyncratic loan-level noise and asks: *is this lender, in this market and time, systematically more expensive than rival lenders?*

2.8 Delinquency and borrower-risk measure

To test whether advertisers simply lend to worse unobserved borrowers, the paper defines delinquency as the share of loans that become 90 or more days delinquent within two years of origination.

The average delinquency rate in ARM loans is above 20%, even during the housing boom.

Interpretation. If advertisers are just pricing higher default risk that is invisible to the econometrician, their local advertising should predict *more* delinquency. The paper finds the opposite sign.

2.9 Craigslist as an instrument for advertising

Craigslist entered different local markets at different times. Mortgage-related classifieds represented over 8% of all financial-services posts on Craigslist, so Craigslist entry plausibly reduced the payoff to paid local mortgage advertising, especially newspaper classifieds.

The first-stage variable is an indicator for whether Craigslist had entered market l by time t .

Interpretation. Craigslist is useful because it perturbs advertising intensity without obviously changing the composition of borrowers in the same way. This creates plausibly exogenous variation in paid advertising.

2.10 Advertising-content measures

The paper complements the price analysis with content analysis of 37,432 print and direct-mail advertising campaigns.

The key content variables are:

- whether the ad contains ARM-related terms,
- whether it mentions one or two rates,
- whether the second rate is an APR,
- whether it uses attention-grabbing teaser phrases such as “as low as,” “intro,” “initial,” or “starting,”
- whether it discloses mortgage horizon, dollar amount, or other quantitative information.

Interpretation. These measures test whether advertising looks informative in substance, not just whether it correlates with pricing outcomes.

3 Models and empirical specifications (all equations + interpretation)

3.1 Competing mechanisms: informative advertising versus persuasion

The paper contrasts two broad theories.

- **Informative advertising:** advertising helps consumers find lower-priced or better-matched mortgages.
- **Persuasive advertising:** advertising draws attention to salient teaser features and steers consumers into expensive contracts.

The main empirical implication is about the sign of the relation between advertising and mortgage expensiveness. Informative advertising predicts that more advertising should be associated with cheaper or equally priced mortgages. Persuasion predicts the opposite.

3.2 Baseline mortgage-pricing equation

The first-step loan-level pricing equation is

$$y_{ijlt} = \beta i_{ijlt} + \alpha_t + \alpha_l + \Gamma X_{ilt} + \varepsilon_{ijlt}. \quad (4)$$

For FRMs, the paper uses the mortgage interest rate as the dependent variable because there is no separate reset rate.

Interpretation. This equation removes the part of mortgage pricing that can be attributed to observables and common market-time factors. What remains is the relative overpricing or underpricing of the contract.

3.3 Why the initial rate is included

The paper explicitly conditions on the initial ARM rate because lenders may have private information about borrower risk that is partly embedded in the teaser rate.

Interpretation. This is an important empirical choice. Without conditioning on the initial rate, differences in reset rates might partly reflect omitted borrower information that lenders used when pricing the mortgage at origination.

3.4 Baseline advertising-expensiveness specification

The core second-stage regression is

$$p_{jlt} = \beta \text{Advertising}_{jlt} + \alpha_j + \alpha_t + \alpha_l + \varepsilon_{jlt}, \quad (5)$$

where Advertising_{jlt} is the total local advertising expenditure of lender j in market l and quarter t .

The preferred specification includes:

- lender fixed effects α_j ,
- quarter fixed effects α_t ,
- region fixed effects α_l .

Interpretation. This is a within-lender, within-region design. It asks whether the same lender becomes more expensive relative to rival lenders in markets where it advertises more.

3.5 Media-specific specification

The paper splits advertising into newspaper advertising and all other advertising:

$$p_{jlt} = \beta_1 \text{NewspaperAdv}_{jlt} + \beta_2 \text{OtherAdv}_{jlt} + \alpha_j + \alpha_t + \alpha_l + \varepsilon_{jlt}. \quad (6)$$

Interpretation. This checks whether the effect is concentrated in the medium most likely to feature salient teaser-rate advertising and most likely to be substituted away from by Craigslist.

3.6 Heterogeneity by borrower sophistication

To test whether less sophisticated borrowers are more susceptible, the paper re-estimates the baseline specification on subsamples split by the demographic composition of the lender's borrower pool:

$$p_{jlt} = \beta \text{Advertising}_{jlt} + \alpha_j + \alpha_t + \alpha_l + \varepsilon_{jlt}, \quad (7)$$

separately for high-versus-low minority share, high-versus-low education share, and high-versus-low poverty share.

Interpretation. The question is not whether minorities or the poor are always charged more. It is whether advertising is *more effective at steering borrowers* in markets where the borrower base is more likely to be vulnerable.

3.7 Delinquency specification

The borrower-risk test uses

$$Delinquent_{jlt} = \beta Advertising_{jlt} + \alpha_j + \alpha_t + \alpha_l + \varepsilon_{jlt}. \quad (8)$$

Interpretation. If the main result is really about worse unobserved borrower quality among advertisers, then advertising should positively predict later delinquency. The paper finds that the coefficient is slightly negative instead.

3.8 Craigslist first stage

The first stage for the IV design is

$$Advertising_{jlt} = \beta_1 AfterCraigslistEntry_{lt} + \alpha_j + \alpha_t + \alpha_l + \nu_{jlt}. \quad (9)$$

Interpretation. Craigslist entry should reduce paid mortgage advertising because online classified ads are free and especially substitute for newspaper classifieds.

3.9 Craigslist second stage

The second stage is

$$p_{jlt} = \beta \widehat{Advertising}_{jlt} + \alpha_j + \alpha_t + \alpha_l + \varepsilon_{jlt}. \quad (10)$$

Interpretation. This is the cleanest causal specification in the paper. It asks whether the part of advertising shifted by Craigslist entry affects mortgage expensiveness.

3.10 Timing specification around Craigslist entry

To examine pre- and post-trends, the paper estimates an event-study style specification with dummies for quarters relative to Craigslist entry:

$$Advertising_{jlt} = \sum_{k=-2}^{2+} \beta_k AfterCraigslistEntry_{lt}^k + \alpha_j + \alpha_t + \alpha_l + \nu_{jlt}, \quad (11)$$

and runs similar regressions with borrower observables as dependent variables.

Interpretation. The point is to check both instrument relevance and instrument validity: advertising should fall around entry, but borrower observables should not shift in a way that suggests Craigslist is proxying for changing borrower quality.

3.11 APR regression

To test whether advertised APR helps consumers find inexpensive lenders, the paper estimates

$$p_{jlt} = \beta APR_{jlt} + \alpha_t + \alpha_l + \alpha_j + \varepsilon_{jlt}. \quad (12)$$

Interpretation. If advertised rates help consumers comparison-shop, lower advertised APRs should map into cheaper lenders. The paper finds that once region and lender fixed effects are included, the correlation is weak and statistically insignificant.

3.12 Standard errors and robustness choices

- In the main lender-DMA-quarter regressions, standard errors are clustered by quarter.
- In the APR regressions, standard errors are clustered by lender.
- The paper also checks robustness to nonlinear first-step pricing models, finer geographic controls, and alternative ways of binning initial rates.

Interpretation. The authors spend considerable effort showing that the measure of mortgage expensiveness is not an artifact of the particular linear specification used to construct it.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the baseline estimate

The baseline estimate is identified from variation in how much the *same lender* advertises across markets and time, controlling for lender fixed effects, region fixed effects, and quarter fixed effects.

Interpretation. This rules out simple stories that some lenders are always predatory, some markets are always expensive, or some quarters are generally both high-advertising and high-price periods.

4.2 Why reverse causality and local-boom stories are weaker than they first appear

A natural concern is that lenders advertise more exactly where they can charge more because local housing demand is booming.

The paper's fixed effects address this in two ways:

- region fixed effects remove persistent cross-market differences in mortgage pricing,
- quarter fixed effects remove aggregate time-series shifts in pricing and advertising.

Interpretation. What remains is not “advertising is high in California” or “advertising is high in 2005.” It is whether a given lender is more expensive in places where it advertises more relative to its own average and relative to rival lenders in the same place.

4.3 Why lender characteristics are unlikely to drive the result

Because the specification includes lender fixed effects, the estimate is not driven by time-invariant lender characteristics such as:

- brand,
- general aggressiveness,
- servicing practices,
- propensity to securitize,
- general customer friendliness,
- average funding costs.

Interpretation. The paper is not just comparing “bad lenders” with “good lenders.” It is exploiting within-lender variation.

4.4 Unobservable borrower quality

A key alternative explanation is that advertisers attract borrowers with worse unobserved repayment ability. The paper counters this using later delinquency.

Table V shows the correlation between advertising and 90-day delinquency within two years is slightly negative and statistically insignificant, both in the ARM sample and in the all-mortgage sample.

Interpretation. If anything, advertisers appear to attract borrowers who are a bit *more* likely to repay. This means the baseline price effect is unlikely to be explained by unobserved risk and may even understate the true persuasion effect.

4.5 Catering costs and origination fees

Another alternative is that advertisers simply serve costlier borrowers and pass those costs through in mortgage pricing.

Using proprietary fee data from a very large subprime lender, the paper examines:

- application fees,
- underwriting fees,
- processing fees,
- total non-interest origination fees.

Differences by minority status or by advertising periods are economically small, often around 1% to 2% of fee levels, far too small to explain the implied mortgage overpricing magnitudes.

Interpretation. This is an important falsification exercise: the pricing effect is too large to be a simple cost-pass-through story.

4.6 Craigslist as an instrument: relevance

Craigslist entry is a strong first stage. The paper shows that Craigslist entry significantly reduces paid mortgage advertising, with the largest decline in newspaper advertising.

Interpretation. This is exactly what the instrument needs. Craigslist is not just correlated with advertising in theory; it measurably lowers it in the data, especially in the closest substitute medium.

4.7 Craigslist as an instrument: exclusion

The key exclusion restriction is that Craigslist entry changes mortgage pricing only through paid advertising, not because it changes borrower quality or other local mortgage fundamentals in a lender-specific way.

The paper supports this with several facts:

- borrower observables such as FICO, LTV, prepayment-penalty share, and low-documentation share move little around entry,
- the only lead effect is a modest decline in advertising one quarter before entry, consistent with some anticipation,
- lenders that advertise on Craigslist are a tiny share of the sample, and dropping them leaves the results unchanged.

Interpretation. The Craigslist strategy is not perfect, but it is persuasive because the first-stage movement is large while the plausible confounds move little.

4.8 What the paper can and cannot distinguish

The paper can strongly reject simple informative-advertising models, but it cannot fully pin down the exact persuasion microfoundation.

It does *not* fully explain:

- why some lenders advertise and others do not,
- why competition does not immediately eliminate profitable persuasive advertising,
- exactly which borrowers are fully aware of reset rates but still borrow,
- whether advertising changes the decision to take a mortgage at all, rather than just which lender is chosen.

Interpretation. The paper is strongest on a narrower but important claim: advertising is associated with worse mortgage choices, not better search outcomes.

4.9 Relation to the literature

The paper sits at the intersection of three literatures.

- **Household finance and the mortgage crisis:** it connects to work on subprime lending, securitization, and predatory lending.
- **Advertising and persuasion:** it brings the consumer-persuasion framework into mortgage markets.
- **Limited attention and salience:** it fits especially well with theories in which consumers overweight salient features and underweight shrouded ones.

5 Tables and figures

The paper is structured cleanly: first document the scale and geography of advertising, then construct mortgage expensiveness, then relate advertising to pricing, then rule out major alternative explanations, then use Craigslist as an instrument, and finally analyze ad content.

5.1 Table I, Figure 1, and Figure 2: the market is large, local, and newspaper-heavy

Table I
Descriptive Statistics

This table presents summary statistics for the different data sets used in our analysis. Panel A reports the total advertising expenditures in the top five DMAs between 2002 and 2006. Panels B and C present summary statistics for ARM and fixed-rate mortgage loans originated by banks over the sample period. Loan-specific attributes include *Reset Rate*, *Initial Interest Rate*, *Reset Time*, *LTV*, *Loan Amount*, credit score (*FICO*), a *Prepay Penalty* indicator, and a *Loan Doc* indicator. Panels B and C also summarize demographic information of the areas in which these loans were given (% *Minority*, *Median Household Income*, % *Poor*, % *Educated* (college degree), % *Female*, and *Average House Value*).

Panel A: Total Advertising Expenditures in Top Five DMAs (2002 to 2006) (thousands)				
#	DMAs	Total	Newspaper	Spot TV
1	Los Angeles	31,451	25,131	2,702
2	New York	15,331	11,265	458
3	San Francisco	8,101	3,722	727
4	Philadelphia	5,057	3,091	999
5	Chicago	4,924	1,735	1,704

Panel B: ARM Loans			
	Mean	Std. Dev.	N
Reset Rate	8.59	1.59	1,182,080
Initial Interest Rate	7.8	1.32	1,182,080
Reset Time	28.24	6.59	1,182,080
LTV	84.68	8.51	1,182,080
Loan Amount	186,601	107,504	1,182,080
FICO	626	53	1,182,080
Prepay Penalty	0.77	0.42	1,182,080
Loan Doc	0.4	0.49	1,182,080
% Minority	25.74	23.24	1,182,080
Median Household Income	47,241	15,055	1,182,080
% Poor	10.86	7.56	1,182,080
% Educated	14.99	23.24	1,182,080
% Female	51.08	3.08	1,182,080
Average House Value	147,978	77,007	1,182,080

Panel C: FRM Loans			
	Mean	Std. Dev.	N
Interest Rate	9.33	2.15	464,530
LTV	91.55	11.77	464,530
Loan Amount	104,528	101,500	464,530
FICO	640	50	464,530
Prepay Penalty	0.59	0.49	464,530
Loan Doc	0.38	0.48	464,530
% Minority	25.02	22.61	464,530
Median Household Income	48,096	15,582	464,530
% Poor	10.87	7.65	464,530
% Educated	15.37	0.08	464,530
% Female	51	2.09	464,530
Average House Value	153,438	81,787	464,530

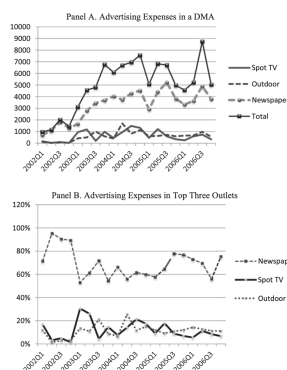


Figure 1. Evolution of mortgage advertising over time. This figure plots the time series of advertising expenditures for the matched lenders over the sample period. Panel A plots the total advertising expenditures and the expenditures in the three outlets over the sample period for our sample of lenders. Panel B provides the time-series evolution of the share of advertising expenditures of the three main advertising channels used by lenders in our sample.

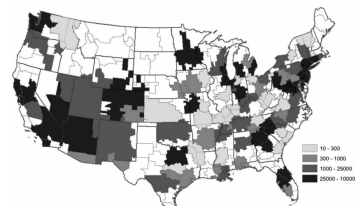


Figure 2. Geographic distribution of DMA-level advertising of mortgage lenders. This map displays the spatial distribution of DMA-level advertising expenditures by mortgage lenders in the 296 DMAs across the United States, over our sample period. Shaded areas represent the total advertising expenditure over our sample period in a given DMA as indicated in the legend (white areas correspond to DMAs for which we do not have advertising information). Advertising numbers in the legend are represented in thousands.

Read:

- Table I shows the matched sample is large: over 1.18 million ARM loans and 464,530 FRM loans. ARM borrowers have average FICO 626 and mean reset rate 8.59%.
- The top five advertising DMAs are Los Angeles, New York, San Francisco, Philadelphia, and Chicago.
- Figure 1 shows total mortgage advertising rises sharply after 2002 and varies substantially over time. Newspapers dominate the mix, although the shares of spot TV and outdoor vary over time.
- Figure 2 shows mortgage advertising is geographically concentrated in major metro areas, but there is still meaningful cross-DMA variation within the national boom.

Economic message. The descriptive exhibits show why the paper can be done at all. Mortgage advertising was both quantitatively important and highly uneven across places, lenders, and media.

5.2 Table II, Figure 3, and Figure 4: there is economically large price dispersion even after controls

Table II
Measuring Mortgage Expensiveness

This table reports estimation results of the following specification: $y_{ijl} = \beta_0 + \beta_1 x_{ijl} + \alpha_i + \alpha_j + \gamma_l + \epsilon_{ijl}$, where i indexes the loan, j indexes the lender, l indexes the quarter, and l indexes the market. The dependent variable is the reset rate on the ARM (first column) or the interest rate on the FRM (second column). *Low Doc* is an indicator for loans that require low documentation at loan application. Other controls include demographic information for the zip codes in which the loans are made (% Minority, median and mean household income, % Poor, % Female, % Education, median and mean house value). Standard errors are clustered by quarter and reported in parentheses under coefficient estimates. ***, **, * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	Y = Reset Rate ARM Loans (1)	Y = Interest Rate FRM Loans (2)
Initial Rate	0.625*** (0.0615)	
Reset Time ($\times 10$)	-0.398*** (0.0347)	
LTV ($\times 10$)	0.108*** (0.0214)	0.006*** (0.0044)
Loan Amount ($\times 10,000$)	-0.0055*** (0.0015)	0.074*** (0.014)
FICO ($\times 100$)	-0.285*** (0.067)	-1.090*** (0.036)
Prepay Penalty	0.196*** (0.029)	-0.210*** (0.055)
Low Doc	0.112*** (0.035)	0.496*** (0.052)
Other Controls	Yes	Yes
Quarter Fixed Effects	Yes	Yes
Region Fixed Effects	Yes	Yes
Observations	1,162,080	464,520
R^2	0.563	0.473

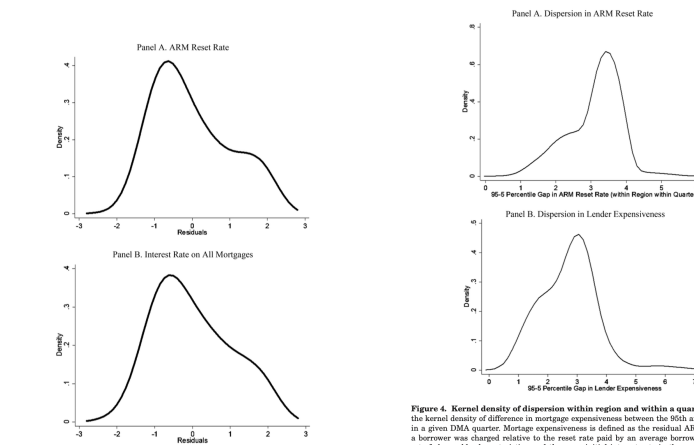


Figure 3. Kernel density of residuals. Panel A plots the kernel density of residual ARM reset rate a borrower was charged. The reset rate residuals are computed from the regression presented in column (1) of Table II. Panel B plots the kernel density of the residual interest rate a borrower was charged. The plotted residuals are the combined ARM and FRM residuals from regressions presented in columns (1) and (2) of Table II.

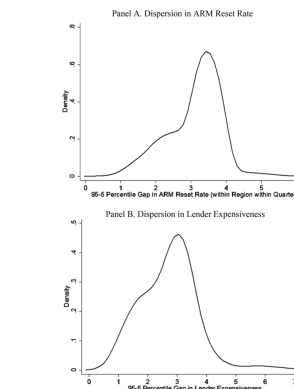


Figure 4. Kernel density of dispersion within region and within a quarter. Panel A plots the kernel density of difference in mortgage expensiveness between the 95th and 5th percentiles in a given DMA quarter. Mortgage expensiveness is defined as the residual ARM reset rate that a borrower was charged relative to the reset rate paid by an average borrower with the same set of observable characteristics, and the same initial interest rate in the same region and the same quarter (the residual from column (1), Table II, averaged at the lender-quarter DMA). Panel B shows the kernel density of the differences between the 95th and 5th percentiles of lender expensiveness in a given area and quarter. Lender expensiveness is computed as the average expensiveness of individual loans issued by the lender for the given location and quarter (combined residuals from columns (1) and (2), Table II, averaged lender-quarter DMA level).

Read:

- Table II estimates the reset-rate and FRM-rate pricing equations. The coefficients have the expected signs: higher FICO lowers rates, higher LTV raises rates, prepayment penalties lower rates, and low-documentation loans are more expensive.
- The pricing equation explains a substantial share of variation: R^2 is 0.563 for ARM reset rates.
- Figure 3 shows a wide distribution of residual mortgage prices after conditioning on observables.
- Figure 4 shows that even after aggregation to the lender level, dispersion remains large: the average 95th–5th percentile lender gap within a DMA-quarter is about 2.8 percentage points.

Economic message. This is the foundation of the paper. If there were no residual pricing dispersion after conditioning on observables, there would be no room to say that some lenders sold worse mortgages than others.

5.3 Figure 5 and Table III: the baseline result is exactly what persuasion predicts

Read:

- Figure 5 shows the distribution of residual reset rates for advertisers is shifted to the right relative to nonadvertisers.
- Table III formalizes this: the coefficient on total advertising is positive and significant in both the ARM sample and the all-mortgage sample.

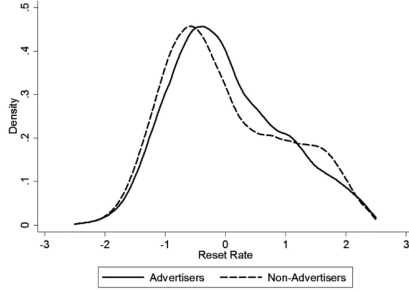


Figure 5. Kernel density plot of residual reset ARM rates for advertisers and nonadvertisers. This figure plots the kernel density of the residual ARM reset rate a borrower was charged. The residual is computed as the reset rate paid by the borrower relative to the reset rate of the average borrower with the same set of observable characteristics and the same initial interest rate in the same region and quarter. We plot the kernel density for lenders who advertise, defined as those with positive advertising expenditures in a given quarter and DMA, and for lenders that do not advertise, defined as those with no advertising expenditures in a given quarter and DMA.

Table III
Advertising and Expensiveness

This table reports the estimation result of the following specification: $p_{jt} = \beta \text{Advertising}_{jt} + \alpha_j + \alpha_t + \alpha_l + \varepsilon_{jlt}$, where the dependent variable measures how expensive mortgages are in a region from a lender. *Lender Expensiveness* is computed by aggregating the individual loan-level residuals obtained from the specification reported in Table II. *Advertising* is the total dollar value of the local advertising of lender j in market l and quarter t . Panel A reports the coefficients using the ARM loan sample. Panel B reports the coefficients using all mortgages. We compute expensiveness for all mortgages using residuals from the specification estimated in column (1) of Table II for ARM loans and in column (2) of Table II for FRM loans. Standard errors are reported in parentheses under coefficient estimates. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Panel A. ARM Loan Sample			
	Y = Lender Expensiveness		
	(1)	(2)	(3)
Advertising (all) ($\times 100$)	0.0314*** (0.0115)		
Advertising (others) ($\times 100$)		0.111 (0.082)	
Advertising (newspapers) ($\times 100$)			0.0368*** (0.0122)
Quarter Fixed Effects	Yes	Yes	Yes
Lender Fixed Effects	Yes	Yes	Yes
Region Fixed Effects	Yes	Yes	Yes
Observations	51,895	51,895	51,895
R ²	0.139	0.139	0.139

Panel B. All Mortgages			
	Y = Lender Expensiveness		
	(1)	(2)	(3)
Advertising (all) ($\times 100$)	0.0239*** (0.009)		
Advertising (others) ($\times 100$)		0.089 (0.072)	
Advertising (newspapers) ($\times 100$)			0.0299*** (0.009)
Quarter Fixed Effects	Yes	Yes	Yes
Lender Fixed Effects	Yes	Yes	Yes
Region Fixed Effects	Yes	Yes	Yes
Observations	51,895	51,895	51,895
R ²	0.140	0.140	0.140

- In Panel A, the coefficient on advertising (all) is 0.0314 and the coefficient on newspaper advertising is 0.0368, both highly significant, while other advertising is not significant.
- The paper interprets the baseline magnitude as roughly 80 basis points for a one-standard-deviation rise in advertising.

Economic message. The central empirical result is not subtle. More local advertising is associated with more expensive mortgages from the same lender in that same local market. That is hard to reconcile with informative-advertising models.

5.4 Table IV, Table V, and Table VI: susceptibility, borrower risk, and cost-based alternatives

Read:

- Table IV shows heterogeneity. Advertising matters in high-minority, low-education, and high-poverty borrower markets, but not in the complementary subsamples.
- Table V shows advertising does not predict more 90-day delinquency. The coefficients are slightly negative and statistically insignificant.
- Table VI uses proprietary fee data from a large subprime lender and shows only small differences in origination-related costs by minority status or by advertising periods.

Economic message. These are some of the sharpest credibility tests in the paper. The result is strongest where persuasion should matter most, but it is not explained by worse hidden borrower quality or by large servicing and origination costs.

Using the IV coefficient, the paper calculates that an average advertising lender's spending implies roughly 181 basis points higher reset rates. Under a conservative assumption that the borrower pays the reset for one year before refinancing, this maps into about \$3,000 in present-value overpayment.

Table VIII shows advertising falls around Craigslist entry, but borrower observables do not move much. The largest, most stable shifts are economically small.

(Continued)

	Y = Lender Expensiveness		
	(1)	(2)	(3)
Advertising (all)	0.078** (0.019)		
Advertising (others)		0.274*** (0.087)	
Advertising (newspapers)			0.131** (0.021)
Quarter Fixed Effects	Yes	Yes	Yes
Lender Fixed Effects	Yes	Yes	Yes
Region Fixed Effects	Yes	Yes	Yes
Observations	50,902	50,902	50,902

	Y = Observables			
	FCO (4)	LTV (5)	Prevalency Penalty (6)	Low Doc. (7)
After Crawlset Entry -2	1.274 (0.827)	-0.0240 (0.120)	0.013 (0.008)	-0.074 (0.008)
After Crawlset Entry -1	-0.538 (0.031)	0.271 (0.164)	0.010 (0.007)	-0.012 (0.001)
After Crawlset Entry 0	0.580 (0.221)	0.124 (0.107)	0.029* (0.011)	-0.017 (0.006)
After Crawlset Entry +1	0.780 (0.076)	0.107 (0.126)	0.011 (0.008)	0.006 (0.004)
After Crawlset Entry +2	0.802 (0.094)	0.144 (0.143)	0.008 (0.017)	-0.009 (0.005)
Quarantined Effects	0.864 (0.086)	0.150 (0.150)	0.012 (0.012)	0.005 (0.005)
Gender Effect	Yes	Yes	Yes	Yes
Gender Effect	Yes	Yes	Yes	Yes
Gender Effect	Yes	Yes	Yes	Yes
Observations	42,452	42,452	42,452	42,452
R ²	0.292	0.233	0.454	0.298

Table IX
Analyzing Advertising Content

This table presents summary statistics on the content of print and direct mail advertising campaigns. In Panel A, we use textual analysis to identify the existence of ARM-related search terms in mortgage advertising campaigns. Search terms include ARM, Reset, and Adjust. Panel B presents results of analysis using advertisements in which two interest rates were advertised. Panel C identifies mortgages that may promote low introductory interest rates (e.g., “as low as,” “intro-,” “initial,” and “starting”). In Panel D, we tabulate the quantitative information disclosed in these campaigns. No Numerical Information refers to campaigns that do mention an actual number. Explicit Interest Rate refers to campaigns that mention a specific interest rate. Below (in reference to a prime rate) refers to campaigns that convey information about interest rate in reference to the prime interest rate offered to high-FICO mortgage borrowers. Dollar Amount refers to advertising campaigns that mention a dollar amount in reference to a downpayment requirement. Horizon refers to advertising campaigns that mention specific loan maturity.

Panel A: ARM-Related Advertising Campaigns	
Search Term	No. Campaigns
ARM	4,238
Reset	13
Adjust	1,885
Explicit Interest Rate	4,234
Total	6,136
Panel B: Advertisement Displays Two Interest Rates	
Second Rate Is APR	11,387
Multiple Products	1,676
Other	128
Total	13,191
Panel C: Phrases on Low Rates	
As Low as	3,632
Intro	769
Initial	346
Starting	608
Total	4,747
Panel D: Quantitative Mortgage Information	
No Numerical Information	6,563
Explicit Interest Rate	26,863
APR Stated	20,146
Below (in reference to a prime rate)	969
Dollar Amount	16,702
Horizon (year, month)	21,992

Table X
APR and Expensiveness

This table reports estimation results of the following specification: $p_{jlt} = \beta APR_{jlt} + \alpha_l + \alpha_t + \alpha_j + \varepsilon_{jlt}$, where the dependent variable measures a lender's mortgage expensiveness in a region. Lender expensiveness is computed by aggregating individual loan level residuals obtained from the specification reported in Table II using the ARM loan sample. *APR* is the advertised APR of lender j in market l and quarter t . Controls are whether the advertisement states a mortgage horizon, reset horizon, and a below-prime initial rate, as well as a count of the terms in the advertisement. Standard errors are clustered by lender and reported in parentheses under coefficient estimates. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	Y = Lender Expensiveness			
	(1)	(2)	(3)	(4)
APR	-0.0205** (0.009)	-0.0258** (0.013)	-0.0105 (0.009)	-0.0093 (0.015)
Controls	No	Yes	Yes	Yes
Quarter Fixed Effects	Yes	Yes	Yes	Yes
Lender Fixed Effects	No	No	No	Yes
Region Fixed Effects	No	No	Yes	Yes
Observations	807	807	807	807
R ²	0.222	0.230	0.421	0.598

Economic message. The content analysis is what lets the paper move from “advertising correlates with higher prices” to “the pattern looks like persuasion.” Reset rates are shrouded, teaser rates are salient, and even the numbers that are disclosed do not help borrowers reliably find inexpensive lenders.

6 Short summary

- The paper asks whether mortgage advertising helps consumers search better or instead steers them into worse mortgage contracts.
- Using matched advertising and mortgage data, it constructs a measure of mortgage expensiveness based on residual reset rates after rich borrower, contract, market, and time controls.
- Lenders that advertise more within a region charge more expensive mortgages in that region.
- This effect is stronger in borrower pools with more minorities, less education, and more poverty.
- Advertisers do not appear to lend to worse unobserved borrowers: delinquency is not higher and may even be a little lower.
- Proprietary fee data suggest catering and origination-cost differences are too small to explain the pricing effects.
- Craigslist entry reduces paid mortgage advertising and instrumented advertising still raises mortgage expensiveness, supporting a causal interpretation.
- Advertising content is consistent with persuasion: teaser rates are salient, reset rates are almost never mentioned, and advertised APRs do not reliably identify cheap lenders.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that local mortgage advertising by subprime lenders is associated with, and plausibly causes, more expensive mortgage choices. Borrowers who respond to advertising do not seem to end up with cheaper lenders; they end up with mortgages that have higher reset rates than those of otherwise similar borrowers.

7.2 Economic intuition

The paper's central economic intuition is that advertising affects what borrowers pay attention to. Mortgage contracts are complex. If the ad makes the introductory rate highly salient and leaves the reset rate in the background, borrowers may search and compare along the teaser dimension rather than along the economically important long-run price dimension. That is exactly the pattern the data support.

7.3 Takeaway

The main takeaway is that in mortgage markets, more advertising did not mean more useful information. It meant more persuasion. This helps explain how household financial mistakes can propagate even in markets for seemingly standardized products, and it gives a microfoundation for why regulators worried so much about mortgage advertising after the crisis.